CURRICULUM VITAE

WON -YOUNG CHOI

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RESEARCH AREAS

Spintronics, Spin-orbit torques, Spin transport, Magnetic materials

EDUCATION

Mar. 2023 – Present
Ph.D. in Material Science and Engineering | Seoul National University, Seoul, Korea
• Advisor: Prof. Hyejin Jang, Co-advisor: Dr. Dong-Soo Han (KIST)

Mar. 2019 – Feb. 2021
M.S. in Physics | Sogang University, Seoul, Korea
• GPA 3.76 / 4.3
• Academic excellence scholarship for freshmen (1st semester, 2019)
• Leader of the thin film subgroup (1st semester, 2020)
• Master thesis was replaced by the published SCI(E) Journal paper (publication No. 11)
• Advisor: Prof. Myung-Hwa Jung

Mar. 2012 – Feb. 2019
B.S. in Physics | Sogang University, Seoul, Korea
• Graduated Summa Cum Laude (3 of 24)
• GPA 3.78/4.3 & Major GPA 4.09/4.3
• Honors scholarship (2nd semester, 2016 & 1st and 2nd semester, 2017)
• Dean's list (overall 2017 semesters)

RESEARCH & WORK EXPERIENCES

Mar. 2023 – Present
Student Researcher | Center for Spintronics, Korea Institute of Science and Technology (KIST), Seoul, Korea

Oct. 2021 – Feb. 2023
Research Intern | Center for Spintronics, Korea Institute of Science and Technology (KIST), Seoul, Korea

Mar. 2021 – Aug. 2021 **Visiting Researcher | Kläui Lab, Johannes Gutenberg-Universität Mainz (JGU)**, Mainz, Germany

Mar. 2019 – Jun. 2020 **Teaching Assistant | Sogang University**, Seoul, Korea
- Experimental Physics 1 class (1st semester, 2019 and 2020)
- Experimental Physics 2 class (2nd semester, 2019)
- Mathematical Physics 1 class (1st semester, 2019)

Mar. 2018 – Feb. 2019 **Undergraduate Student Researcher | Sogang University**, Seoul, Korea
- Fabricated magnetic thin film using magnetron co-sputtering
- Evaluated magnetic properties of magnetic thin film using MPMS-3 (SQUID-VSM)
- Investigated electrical properties of magnetic thin film

Oct. 2013 – Oct. 2015 **Air Materials Supply Sergeant | 15th Special Missions Wing, Republic of Korea Air Force**, Seoul, Korea
- Served military service
- Chief soldier (Jan. 2015 – Jul. 2015)

PUBLICATIONS & PRESENTATIONS

PUBLICATIONS (* co-first author)

1. J.-B. Choi, H. Seong, **W.-Y. Choi**, S.-J. Noh, J.-S. Lee, H.-S. Lee, Y.-C. Kim, O.-J. Lee, B.-C. Min, J.-H. Kwon, D.-S. Han, Spin Hall Magnetic Sensors with Dynamic Offset Calibration and Structurally Tunable Sensitivity. IEEE Sensors Journal **25**, 21, 39605-39611 (2025).

2. **W.-Y. Choi***, J.-H. Ha*, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Magnetization switching driven by magnonic spin dissipation. Nature Communications **16**, 5859 (2025).

3. S. Yu, J. Park, Y. Jo, W. Kim, H. Kim, **W.-Y. Choi**, M. Kim, D.-S. Han, M.-H. Jung, K. Rhie, K. Lee, The control of magnetization reversal by tuning the interlayer exchange interaction in magnetic multilayers. Journal of the Korean Physical Society **86**, 62-67 (2024).

4. H. Seong, **W.-Y. Choi**, J. Choi, D.-H. Kim, T.-E. Park, B.-C. Min, H.-J. Choi, D.-S. Han, Magnetic properties of GdFeCo thin films tailored by sputtering conditions. Current Applied Physics **68**, 131-137 (2024).

5. S. Jena, R. Urkude, **W.-Y. Choi**, K. K. Pandey, S. Karwai, M. H. Jung, J. Gardner, B. Ghosh, V. R. Singh, Electronic Structures of Skyrmionic Polycrystalline MnSi Thin Film Studied by Resonance Photoemission and X-ray Near Edge Spectroscopy. Journal of Applied Physics **135**, 165301 (2024).

6. S. Jena, R. Dawn, **W.-Y. Choi**, Y. Singh, A. Chosh, S. K. Sahoo, M. H. Jung, J. Gardner, V. K. Verma, K. Amemiya, V. R. Singh, Texture, Electronics and Magnetic States of Mn Atoms in Polycrystalline MnSi Thin Films Facing ac-Sapphire Substrate Studied by Soft X-ray Magnetic Circular Dichroism. Thin Solid Films **786**, 140097 (2023).

7. F. Kammerbauer, **W.-Y. Choi**, F. Freimuth, K. Lee, F. Robert, D. -S. Han, H. J. M. Swagten, Y. Mokrousov, M. Kläui, Controlling 3D spin textures by manipulating sign and amplitude of interlayer DMI with electrical current. Nano letters **23**, 15, 7070-7075 (2023), arXiv: 2209. 01450.

8. S. Jena, **W.-Y. Choi**, J. Gardner, M. H. Jung, S K Srivastava, V K Verma, K Amemiya, V R Singh, Evolution of bulk magnetic structure in MnSi thin film: a soft x-ray magnetic circular dichroism study. Physica Scripta **98**, 075927 (2023).

9. **W.-Y. Choi**, W. Yoo, M.-H. Jung, Emergence of the topological Hall effect in a tetragonal compensated ferrimagnet $Mn_{2.3}Pd_{0.7}Ga$. NPG Asia Materials **13**, 79 (2021).

10. **W.-Y. Choi**, J. H. Jeon, H. W. Bang, W. Yoo, SK. Jerng, S. H. Chun, S. Lee, M.-H. Jung, Proximity-Induced Magnetism Enhancement Emerged in Chiral Magnet MnSi/Topological Insulator Bi_2Se_3 Bilayer. Advanced Quantum Technologies **4**, 2000124 (2021).

11. **W.-Y. Choi**, H.W. Bang, S. H. Chun, S. Lee, M.-H. Jung, Skyrmion Phase in MnSi Thin Films Grown on Sapphire by a Conventional Sputtering. Nanoscale Research Letters **16,** 7 (2021).

PRESENTATIONS

1. **W.-Y. Choi**, J.-H. Ha, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Self-generated Spin-orbit Torques Driven by Magnonic Spin dissipation in Antiferromagnet/Ferromagnet Bilayer. 2025 70th MMM Annual Conference Invited Talk.
2. **W.-Y. Choi**, J.-H. Ha, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Magnonic Spin Dissipation Driven Anisotropic Spin Torque in Easy-Axis Antiferromagnet/Ferromagnet Bilayer. 2025 KMS Summer Conference poster presentation.

3. **W.-Y. Choi**, H.-J. Ha, M.-S. Jung, S. Kim, O. Lee, B.-C. Min, M. Kläui, H. Jang, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin-orbit torque by magnon dissipation. 2023 MML Conference poster presentation.

4. **W.-Y. Choi**, H.-J. Ha, H. Jang, B.-C. Min, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin Hall magnetoresistance and unidirectional magnetoresistance in ferromagnet metal/antiferromagnet insulator bilayers. 2023 KMS Summer Conference poster presentation.

5. **W.-Y. Choi**, H.-J. Ha, M.-S. Jung, B.-C. Min, M. Kläui, J.-I. Hong, K.-W. Kim, D.-S. Han, Spin-orbit torque by magnon dissipation. 2022 KMS Winter Conference poster presentation.

6. **W.-Y. Choi**, H. W. Bang, Y. C. Park, M.-H. Jung, Topological Hall effect in a mixed phase of Mn_3Ga thin film. 2022 KMS Summer Conference poster presentation.

7. **W.-Y. Choi**, W. Yoo, M. H. Jung, Observation of unconventional anomalous Hall effect at compensated $Mn_{2.3}Pd_{0.7}Ga$ thin film. 2020 KPS Fall Meeting oral presentation.

8. **W.-Y. Choi**, W. Yoo, H.W. Bang, M.T. Park, K. Lee, M.-H. Jung, Compensated ferrimagnetism of $Mn_{2+\delta-x}Pd_xGa$ thin film. 2019 KPS Fall Meeting poster presentation.

HONORS & AWARDS

Jul. 2026	**STAR Outstanding Graduate Student Award,** The Korean Magnetics Society (KMS) 2026
Dec. 2025	**PSI Excellent Student Researcher Award,** Korea Institute of Science and Technology Post-Silicon Semiconductor Institute 2025
Oct. 2025	**Condensed Matter Physics MSM Awards,** The Korean Physical Society (KPS) 2025 Fall Meeting
Nov. 2022	**Best Poster Award,** The Korean Magnetics Society (KMS) 2022 Winter Conference
May. 2022	**Poster Award,** The Korean Magnetics Society (KMS) 2022 Summer Conference
Jan. 2021	**International Academic Papers Scholarship ($500)**, Ministry of Education, Korean government
Oct. 2020	**Outstanding Presentation Award**, The Korean Physical Society (KPS) 2020 Fall Meeting
Mar. 2019	**Academic Excellence Scholarship for Freshmen ($3200)**, Graduate School of Sogang University, Seoul, Korea

Mar. 2018	**CHO-LEE Scholarship ($1000)**, 1st semester, 2018, Corporation CHO-LEE, Seoul, Korea
Jan. 2018	**Dean's List Award**, Sogang University, Seoul, Korea (Award for Top 3% Student)
Sept. 2016 – Dec. 2017	**Honors Scholarship ($2100)**, 2nd semester, 2016, and 1st & 2nd semester, 2017, Sogang University, Seoul, Korea

PERSONAL SKILLS

• **Sample Preparation Techniques:** Magnetron Sputtering, Photolithography, Ar ion milling, Wire Bonder

• **Measurements Techniques:** Scanning Electron Microscopy (SEM), Energy dispersive X-ray spectroscopy (EDXS), X-ray diffraction (XRD), Van der Pauw Method, Magnetic Properties Measurement System (MPMS), Physical Properties Measurement System (PPMS), Electrical Transport Properties Measurement, 2nd Harmonics Spin-orbit torque Measurement

• **Programming Language:** C, Python, Labview

• **Software:** Origin Lab, Qtiplot, VESTA, ElmerFEM, NanoCAD, FreeCAD, Illustrator, Blender

• **Languages:** Native Korean, Fluent English, Intermediate Japanese

• **Sports:** Kendo (2nd dan)